

Problem 15.2

The annual payment on a house is \$18,000. If payments are made for 40 years, how much is the house worth assuming annual interest rate of 6%?

Solution

Let X be the present value of the house. Since the payments are annual and are made for 40 years, we value the stream of payments as an annuity-immediate with annual effective interest rate

$$i = 0.06.$$

Therefore,

$$X = 18000a_{\overline{40}|}.$$

Using the present value formula for an annuity-immediate,

$$a_{\overline{40}|} = \frac{1 - v^{40}}{i},$$

where

$$v = \frac{1}{1 + i} = \frac{1}{1.06}.$$

Thus,

$$X = 18000 \left(\frac{1 - (1.06)^{-40}}{0.06} \right).$$

Finally,

$$X \approx 270833.3437.$$

Hence, the house is worth approximately

$$\boxed{\$270,833.34}.$$